

When Synthetic Telecom Benchmarks Miss Controlled-Real 5G Faults: Origin-Aware Auditing and Calibration

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ABSTRACT

Reliable anomaly detection is a prerequisite for 5G operations and self-optimizing 6G, but real labeled telecom faults are rare and difficult to release, so anomaly detectors are typically evaluated using injected or generated synthetic anomalies. We do not claim that all synthetic telecom anomalies fail; we ask, *for a given benchmark*, whether its synthetic anomalies cover the operating regime occupied by controlled-real faults, and release *origin-aware benchmark auditing* as a deployment-oriented evaluation procedure that treats anomaly origin as a first-class variable. On two independent public 5G/Open RAN testbed corpora (TelecomTS and SpotLight) whose dominant separating KPI families differ – RSRP-mediated and RLC-timer-mediated respectively – the audit exposes the same operationally consequential failure that aggregate F_1 hides: across six detector families, synthetic-only training preserves $> 90\%$ recall on perturbation-synthetic anomalies but collapses to 0–7.7% on the controlled-real fault at the natural anomaly rate. We localize the failure to an *operating-regime coverage gap* rather than a feature-leakage artifact: it persists when all RSRP-derived features are removed, replicates on the second corpus with a different separating KPI family, and is not closed by importance-weighting or uL-SIF density-ratio adaptation at the matched budget. The audit then estimates a corpus-specific calibration budget; 18 controlled-real windows on TelecomTS lift recall to 0.92, while matched-budget extra synthetic, extra normal, and reweighting controls leave recall near zero. Under aggregate F_1 the synth-only detector would be selected; under the audit it is rejected. We release a reproducible toolkit (14 notebooks, ~30-minute laptop reproduction) and an operator checklist for pre-deployment benchmark validation, with industrial validation as a shadow-mode CI gate inside Nokia’s Agentic Anomaly Detection pipeline.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Information systems** → **Data mining**.

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KEYWORDS

5G, anomaly detection, benchmark analysis, distribution shift, real-synthetic gap, telecom observability, deployed systems, applied research, time series

1 INTRODUCTION

Reliable anomaly detection is a core requirement for 5G operations and an increasingly important prerequisite for autonomous 6G networks. Modern RAN deployments expose high-dimensional KPI streams that can reveal radio interference, congestion, software faults, mobility failures, and resource-allocation bugs, but these signals are useful only if detectors trained offline transfer to the operating regimes that appear in deployment. Real telecom failures are rare, operator datasets are difficult to release, and expert labeling is expensive, so many telecom anomaly studies evaluate detectors using injected, generated, or simulated anomalies [13, 29, 32]. These choices are practical, but they leave a benchmark assumption largely untested: that synthetic anomalies cover the operating regime occupied by controlled-real RF faults in public 5G/Open RAN testbeds.

We do not claim that all synthetic telecom anomalies fail. We ask, *for a given benchmark*, whether its synthetic anomalies cover the operating regime occupied by controlled-real faults – and we provide an audit that detects when they do not, before model selection. We study this on two independent public 5G/Open RAN testbed corpora whose dominant separating KPI families differ: TelecomTS [8] (controlled-real Jamming alongside ten perturbation-synthetic types; separator is RSRP) and SpotLight [28] (controlled-real RADIO interference alongside perturbation-synthetic PDCP/MAC/NETWORK anomalies; separator is the layer-2 RLC-timer family). Throughout, *controlled-real* means an anomaly injected through RF or hardware mechanisms and observed through a real RAN stack, but *not* a passively collected field outage from a deployed operational network. Some of these gaps arise from collection protocols (e.g. UE placement during fault campaigns) rather than from synthetic-anomaly generators themselves; the audit’s value is to expose either kind of gap before model selection. The audit has two linked effects: a distributional gap (origins occupy different KPI operating regimes) and a transfer gap (a detector trained without controlled-real anomalies performs well on perturbation-synthetic anomalies

while missing controlled-real ones), and aggregate benchmark scores can remain high even when the second has not been learned.

We argue that anomaly origin should be treated as a first-class benchmark variable in operational anomaly detection evaluation, and we operationalize this view as an applied audit workflow with a released, reproducible toolkit and pre-production integration into Nokia's in-house anomaly-detection platform. Our contributions are:

- **Origin-aware benchmark auditing** as a deployment-oriented evaluation framework. The audit treats anomaly origin (controlled-real vs perturbation-synthetic) as a first-class variable and tests whether synthetic anomalies cover the operating regime occupied by controlled-real testbed faults, using KPI-level statistical tests, classifier two-sample tests [18], MMD [12], train-on-synthetic/test-on-real protocols, and calibration sweeps.
- **Aggregate metrics can hide deployment failure.** Across six detector families (tabular ensembles, frozen foundation-model encoders, and end-to-end deep classifiers), synthetic-only training preserves $> 90\%$ synthetic-anomaly recall but collapses to 0–7.7% on the controlled-real fault at the natural anomaly rate, while aggregate F_1 remains visually acceptable.
- **The failure is an operating-regime coverage gap, not a feature-leakage artifact.** It persists when all RSRP-derived features are removed (§4.2), is not explained by recorded deployment context, and replicates on the independent SpotLight Open RAN corpus [28] where the dominant separating KPI family differs.
- **Small real calibration repairs the gap; matched-budget synthetic alternatives do not.** We frame this as a per-corpus calibration-budget estimation procedure rather than a universal number: $f \approx 10\%$ of the train-eligible controlled-real pool suffices on TelecomTS, while SpotLight requires $f \approx 25\%$.
- **Released audit toolkit with measured utility.** 14 reproducible notebooks plus a shared module regenerate every table and figure on a consumer laptop in ~ 30 min, and ship with an operator pre-deployment checklist for benchmark validation. We do not propose new statistical machinery: we operationalize a deployment-time audit workflow that changes operational model-selection decisions, and we validate that workflow on two public corpora and one in-house Nokia pipeline (§5).

2 RELATED WORK

Telecom benchmarks and synthetic anomalies. TelecomTS provides absolute-scale 5G KPI traces with both controlled Jamming and synthetic anomaly types [8]; SpotLight studies explainable anomaly detection on an Open RAN testbed with mixed RADIO/PDCP/MAC/NETWORK anomalies [28]; CellPAD [32], TSAGen [29] and Simba [13] similarly rely on injected, generated, or simulated anomalies in cellular KPI traces; and BOOM/Toto broadens observability benchmarking to large-scale production metrics [7]. These works motivate controlled and synthetic anomaly generation; our contribution is to *measure* when such anomalies fail to transfer to controlled-real testbed faults, and to release the audit workflow that does so.

Anomaly detection and protocol auditing. A long line of work develops KPI and multivariate anomaly detectors for operational systems [2, 16, 25, 26, 33, 34], and recent time-series foundation models have become common baselines for downstream tasks [1, 7, 9,

11]. A complementary line audits *evaluation*: point-adjustment can inflate TSAD scores [15], simple baselines can match deep detectors [3], one-liner baselines can match foundation-model anomaly scores under fixed protocols [35], and benchmarking efforts such as TimeEval/TSB-AD push reproducible evaluation [17, 30]. These audits assume a fixed label set; we audit a different layer of the benchmark — whether labels from different *anomaly origins* are valid proxies for the controlled-real population — which is complementary to metric-reliability auditing.

Distribution shift, domain adaptation, and subgroup failure. Using the standard dataset-shift taxonomy [19, 23], the real–synthetic gap is *origin-conditional covariate shift*: controlled-real and perturbation-synthetic anomaly windows have different $P(X | \text{ORIGIN})$ though both are labeled anomalous, so detectors that learn a single aggregate anomaly distribution can fail when the controlled-real origin occupies a different KPI operating regime. We use classifier two-sample tests [18] and kernel MMD [12] as diagnostics, not as new detection methods, and compare against importance-weighted training and uLSIF-style covariate-shift correction [5, 14, 27] at the same effective sample budget in §4.3 (Table 5). The intellectual neighbourhood also includes work on shortcut learning [10], hidden stratification in clinical ML [21], and dataset-audit / benchmark-artifact studies [24]: origin-aware auditing operationalises subgroup-failure detection for the specific case where the subgroup variable is *anomaly origin* and the consumer is a deployment-time model selector.

3 BENCHMARK AND AUDIT PROTOCOL

Anomaly-origin terminology. In this paper, **controlled-real** means an anomaly injected through RF or hardware mechanisms and observed through a real RAN stack in a testbed (TelecomTS Jamming, SpotLight RADIO). **Perturbation-synthetic** means an anomaly generated by perturbing KPI traces (the ten remaining TelecomTS types; SpotLight PDCP/MAC/NETWORK). **Field-real** would mean a passively observed outage from a deployed operational network; neither corpus contains field-real anomalies, and our conclusions should not be over-extrapolated to that regime without additional data.

Corpora. TelecomTS [8] contains 32,000 windows (30,765 normal, 1,235 anomalous; 11 anomaly types) of 128 timesteps over 16 numeric KPI channels covering radio/PHY (RSRP, UL_SNR¹), error rates (DL/UL_BLER), modulation/coding (DL/UL_MCS), scheduler/resource (PRB utilisation, current PRBs, UL_NPRB, estimated UL_NPRB), and traffic counters (TX/RX bytes and packet counts). SpotLight [28] is a public Open RAN dataset from a different 5G testbed with 241 controlled-real RADIO windows and 436 perturbation-synthetic windows over 452 RAN/platform channels. TelecomTS metadata also includes deployment context (zone, application, mobility, congestion); the audit found that controlled-real Jamming

¹**KPI primer.** RSRP = Reference Signal Received Power (downlink signal-quality indicator, dBm). UL_SNR = Uplink Signal-to-Noise Ratio. DL/UL_BLER = Block Error Rate per direction. DL/UL_MCS = Modulation & Coding Scheme index. PRB = Physical Resource Block (RAN scheduler unit). RLC (used on SpotLight, §4.4) = Radio Link Control, layer-2 protocol counters.

Table 1: Partitions used in the TelecomTS audit.

Partition	Primary use	Train/Val/Test	Anom. rate
Controlled-500	Origin transfer	300/100/100	50%
Balanced det.	KPI sensitivity	1580/396/494	50%
Full-scale	TSTR and calibration	25,600/–/6,400	3.9%
Context LOO	Coverage/split sens.	all other/–/held-out	variable

appears only in Zone A and that mobile / congestion-positive windows contain no anomalies, so context experiments serve as coverage/split-sensitivity audits rather than balanced benchmarks. All detection audits use only numeric KPI channels.

Features and primary diagnostic detector. We represent each window with 240 handcrafted numeric summaries (15 statistics per KPI: level, dispersion, trend, change-energy, autocorrelation, and spectral descriptors), used for per-KPI tests, C2ST [18], MMD [12], train-on-synthetic/test-on-real (TSTR) detection, calibration, and single-KPI sanity checks. The primary diagnostic detector is HistGradientBoostingClassifier [22] on these summaries; we deliberately do not introduce a new detector architecture. Model-family robustness is established by repeating the TSTR protocol with five additional detectors (§4.1).

Splits. We use task-specific partitions (Table 1). The *controlled-500* split isolates transfer by balancing controlled-real and perturbation-synthetic counts in the test set; the *balanced-detection* split supports KPI sensitivity at 50% rate; the *full-scale* split preserves the natural 3.9% rate and is used for the main TSTR and calibration experiments (all controlled-real Jamming windows excluded from synthetic-only training); the *context LOO* partitions hold out one context value at a time. Thresholds are tuned on a disjoint validation subset carved from training (max- F_1) and then applied unchanged on the held-out test set. Because threshold choice can hide origin-specific failures, we report subgroup recalls and AUROC alongside aggregate F_1 . All splits are per-window random shuffles with seed=42 over the windowed corpora released by [8, 28]; windows are pre-extracted with fixed non-overlapping boundaries, so per-window shuffling does not introduce temporal leakage between adjacent train and test windows. The controlled-real test pool (69 Jamming windows on TelecomTS / 35 RADIO windows on SpotLight) is fixed once and is disjoint from the calibration pool across every sweep value f and every seed. Per-trace / per-session group-aware splits would tighten this further; the absence of session ids in the released TelecomTS metadata is a limitation we note in §6.

4 DIAGNOSING AND REPAIRING THE REAL–SYNTHETIC GAP

The audit measures two linked effects: (i) a *distributional gap* between controlled-real and perturbation-synthetic anomaly windows in KPI-summary space, and (ii) a *transfer gap* in which models trained without controlled-real anomalies fail to detect them at test time. Aggregate binary F_1 can remain high even when the controlled-real origin is missed entirely.

4.1 Synthetic-Only Training Fails Across Detector Families

Table 2 evaluates the transfer gap across six representative detector families covering tabular ensembles, frozen time-series foundation encoders, and supervised end-to-end deep classifiers, so the conclusion does not depend on a single architecture: HistGradientBoosting, RandomForest, and XGBoost on the 240 engineered KPI summaries [6, 22]; a frozen Toto encoder with a small binary head [7]; and end-to-end TimesNet and PatchTST on the raw 128×16 windows [20, 31]. The full 11-detector table (including LR, kNN, MLP, frozen Mantis, and InceptionTime) is reproduced in the released artifact and shows the same pattern.

The *controlled-500* split isolates training composition by balancing controlled-real and perturbation-synthetic counts in the test set. With both origins in training, every detector recognises both. With controlled-real Jamming removed, perturbation-synthetic recall remains substantial but controlled-real recall drops sharply for the tabular and frozen-backbone models and becomes unstable for the raw-window deep models. The decisive evidence comes from the *full-scale* split at the natural anomaly rate: every synthetic-only detector misses or nearly misses controlled-real Jamming while aggregate F_1 and AUROC remain visually acceptable because perturbation-synthetic anomalies are still recognised. Origin-aware reporting is necessary to expose this failure mode.

4.2 The Failure is an Operating-Regime Coverage Gap

We next show that synthetic-only training fails because the perturbation-synthetic anomalies do not populate the KPI operating regime occupied by the controlled-real RF fault. Across all 1,235 anomalous windows, every numeric KPI mean differs significantly between origins under Kolmogorov–Smirnov tests after Benjamini–Hochberg correction, with RSRP the strongest separator (controlled-real Jamming mean RSRP ≈ -76 dB vs synthetic ≈ -109 dB; KS = 1.0; Cohen’s $d = 3.17$). A 5-fold C2ST on the 240 engineered summaries distinguishes the two origins at 100% accuracy and AUC 1.000; an RBF-kernel MMD test rejects equality at $50.9\times$ the permutation-null mean ($p = 0.000999$). Per-anomaly-type RSRP distributions (Fig. 1) localise the gap: the testbed’s standard operating point is at low RSRP because the UE was usually placed away from the RU, so the Normal and the ten perturbation-synthetic types all cluster near -110 dBm; controlled-real Jamming was instead collected with the UE close to the RU (Zone A only), at mid-cell RSRP. The synthetic-only training signal therefore lacks the high-RSRP regime in which the controlled-real fault appears.

The gap is not a trivial feature-leakage artifact. We evaluate three follow-up tests to assess the leakage interpretation. First, on context-matched comparisons (synthetic anomalies with the same zone, application, mobility, and congestion metadata as the controlled-real Jamming windows) the gap persists (RSRP KS = 1.0, Cohen’s $d = 4.29$), so recorded deployment context does not explain it. Second, even after removing all RSRP-derived features the C2ST remains above 99% accuracy and synthetic-only training continues to miss controlled-real Jamming; the same $f = 10\%$ calibration budget that fully repairs the full-feature model (0.91 controlled-real

Table 2: Multi-detector transfer audit (mean $\pm$ std over 10 seeds). Synthetic-only training collapses controlled-real Jamming recall to $\leq 5\%$ across all six families on Full-6.4k while aggregate F_1 and AUROC remain visually acceptable. The full 11-detector version (LR, kNN, MLP, frozen Mantis, InceptionTime) reproduces this pattern and is in the released artifact. std=0 rows are bit-for-bit deterministic given cached frozen embeddings and a convex LR head.

Detector	Ctrl-500 / all				Ctrl-500 / synth				Bal-494 / synth				Full-6.4k / synth			
	F1	AUC	Real	Synth	F1	AUC	Real	Synth	F1	AUC	Real	Synth	F1	AUC	Real	Synth
<i>Tabular detectors on 240 engineered KPI summaries</i>																
HGB	0.95 $\pm$ 0.02	0.99 $\pm$ 0.01	100 $\pm$ 0%	91 $\pm$ 4%	0.65 $\pm$ 0.03	0.75 $\pm$ 0.06	8 $\pm$ 8%	92 $\pm$ 4%	0.85 $\pm$ 0.00	0.88 $\pm$ 0.00	12 $\pm$ 0%	99 $\pm$ 0%	0.81 $\pm$ 0.01	0.88 $\pm$ 0.01	0 $\pm$ 0%	96 $\pm$ 1%
RF	0.94 $\pm$ 0.02	0.99 $\pm$ 0.01	100 $\pm$ 0%	90 $\pm$ 7%	0.66 $\pm$ 0.04	0.80 $\pm$ 0.04	16 $\pm$ 13%	92 $\pm$ 4%	0.84 $\pm$ 0.00	0.92 $\pm$ 0.00	17 $\pm$ 4%	97 $\pm$ 0%	0.80 $\pm$ 0.00	0.90 $\pm$ 0.01	0 $\pm$ 0%	94 $\pm$ 1%
XGB	0.94 $\pm$ 0.01	0.98 $\pm$ 0.01	100 $\pm$ 0%	93 $\pm$ 4%	0.64 $\pm$ 0.06	0.73 $\pm$ 0.06	10 $\pm$ 10%	90 $\pm$ 7%	0.83 $\pm$ 0.01	0.87 $\pm$ 0.00	9 $\pm$ 5%	97 $\pm$ 1%	0.82 $\pm$ 0.00	0.83 $\pm$ 0.01	0 $\pm$ 0%	97 $\pm$ 1%
<i>Modern supervised detectors on raw 128 $\times$ 16 KPI windows</i>																
Toto (frozen)	0.74 $\pm$ 0.07	0.82 $\pm$ 0.07	83 $\pm$ 13%	78 $\pm$ 12%	0.53 $\pm$ 0.06	0.68 $\pm$ 0.04	18 $\pm$ 7%	72 $\pm$ 13%	0.76 $\pm$ 0.01	0.83 $\pm$ 0.00	23 $\pm$ 5%	90 $\pm$ 3%	0.59 $\pm$ 0.01	0.88 $\pm$ 0.00	0 $\pm$ 0%	68 $\pm$ 4%
TimesNet (e2e)	0.65 $\pm$ 0.02	0.57 $\pm$ 0.07	92 $\pm$ 12%	92 $\pm$ 12%	0.58 $\pm$ 0.10	0.55 $\pm$ 0.07	67 $\pm$ 24%	74 $\pm$ 23%	0.65 $\pm$ 0.01	0.66 $\pm$ 0.01	62 $\pm$ 7%	84 $\pm$ 4%	0.36 $\pm$ 0.03	0.76 $\pm$ 0.03	0 $\pm$ 0%	39 $\pm$ 7%
PatchTST (e2e)	0.67 $\pm$ 0.02	0.66 $\pm$ 0.06	93 $\pm$ 8%	92 $\pm$ 7%	0.58 $\pm$ 0.13	0.62 $\pm$ 0.08	60 $\pm$ 34%	71 $\pm$ 24%	0.71 $\pm$ 0.01	0.78 $\pm$ 0.01	20 $\pm$ 5%	90 $\pm$ 3%	0.64 $\pm$ 0.01	0.81 $\pm$ 0.01	0 $\pm$ 1%	73 $\pm$ 3%

recall, Table 3) only *partially* repairs the no-RSRP and no-level subsets (0.28 and 0.11 recall respectively, both significantly above the $f=0$ baseline of ≤ 0.004 but well below the 0.91 of the full-feature model). The audit’s diagnosis is therefore: RSRP is the dominant single signal *and* the dominant transferable signal — when it is removed, the synthetic-only training pool is missing both at once, so a fixed 18-window budget is below the per-feature-subset budget the audit estimates, and the operationally correct action for a practitioner without RSRP is the audit’s verdict (refuse certification) followed by a per-subset calibration sweep rather than a fixed prescription. Third, the same operating-regime coverage gap replicates on the independent SpotLight corpus (§4.4) where the dominant separating KPI family is layer-2 RLC-timer counters rather than RSRP. The underlying issue is broader: *synthetic perturbations do not cover the KPI operating regime occupied by the controlled-real RF fault, and the audit discovers this coverage failure before model selection.*

*Is the calibrated detector just learning “high RSRP”? A natural reviewer concern is that the $f=10\%$ calibration recipe could be exploiting a UE-close-to-RU shortcut: if controlled-real Jamming lives at high RSRP and only at high RSRP, the calibrated model might flag any high-RSRP Normal window as anomalous. We test this directly by stratifying held-out Normal windows by per-window mean RSRP and reporting the calibrated detector’s per-bin FPR (Table 4). The calibrated FPR stays $\leq 0.44\%$ in every populated bin, including the highest-RSRP bin (-85 to -75 dBm) immediately adjacent to where controlled-real Jamming lives, so the calibrated detector is *not* simply flagging high RSRP. The bin ≥ -75 dBm is empty: TelecomTS collected no Normal traffic with the UE that close to the RU, which is itself a collection-protocol artifact we discuss next and a coverage limitation the audit reports rather than papers over.*

A note on collection vs methodology. The TelecomTS operating-regime gap is, at its mechanistic root, a *collection-protocol* artifact: controlled-real Jamming was collected with the UE close to the RU (Zone A only) while perturbation-synthetic anomalies were injected on top of traces collected at the standard testbed operating point. This shifts how the methodological claim should be read: the audit’s value is to expose such artifacts before model selection,

Table 3: Anti-confound transfer ablation (HGB, full-scale split, mean $\pm$ std over 10 seeds). Synthetic-only training fails on controlled-real Jamming in every feature subset, including the subset with all RSRP-derived features removed and the subset with all absolute-level statistics removed. A fixed $f=10\%$ (18 windows) calibration budget *fully* repairs the gap on the full feature set (controlled-real recall 0.91); on the RSRP-removed and level-removed subsets the same budget still improves recall significantly over the $f=0$ baseline (from ~ 0 to 0.28 / 0.11) but does not close it, indicating that 18 windows is below the per-subset budget the audit estimates rather than that the failure is RSRP-shortcut. The failure is an operating-regime coverage gap, not a trivial RSRP-feature shortcut: it persists across feature subsets and replicates on SpotLight where the dominant separator is RLC-timer rather than RSRP (§4.4).

Feature subset	Calib.	Real	Synth	F_1	AUC
All 240 feats	$f=0$	0.001 $\pm$ 0.005	0.957 $\pm$ 0.016	0.810 $\pm$ 0.009	0.882 $\pm$ 0.009
All 240 feats	$f=10\%$	0.907 $\pm$ 0.063	0.966 $\pm$ 0.011	0.962 $\pm$ 0.009	0.998 $\pm$ 0.001
No RSRP (225 feats)	$f=0$	0.000 $\pm$ 0.000	0.943 $\pm$ 0.011	0.807 $\pm$ 0.005	0.904 $\pm$ 0.010
No RSRP (225 feats)	$f=10\%$	0.281 $\pm$ 0.126	0.960 $\pm$ 0.007	0.848 $\pm$ 0.017	0.974 $\pm$ 0.008
No level (176 feats)	$f=0$	0.004 $\pm$ 0.007	0.952 $\pm$ 0.007	0.802 $\pm$ 0.006	0.893 $\pm$ 0.008
No level (176 feats)	$f=10\%$	0.112 $\pm$ 0.059	0.948 $\pm$ 0.010	0.814 $\pm$ 0.012	0.950 $\pm$ 0.012

which is exactly the case where standard aggregate-metric reporting conceals them. Practitioners typically cannot retroactively redesign collection on top of an existing benchmark, and operational telecom benchmarks frequently have coverage imbalances of this kind. SpotLight’s RLC-timer gap (§4.4) is mechanistically different from RSRP yet yields the same operator-decision change, which is why we report this work as a cross-testbed replication of the methodology rather than a single-corpus result.

4.3 Calibration Budget Estimation

The audit then estimates a *per-corpus calibration budget*: how many controlled-real windows must be added to synthetic-only training before transfer is reliable. We do not claim a universal number; we ship the procedure. Let $f \in [0, 1]$ denote the fraction of the train-eligible controlled-real Jamming pool added to synthetic-only training; the TelecomTS pool contains 185 windows, so $f =$

Table 4: High-RSRP Normal-FPR stratified control on TelecomTS (HGB, $f = 10\%$ controlled-real calibration, 10 seeds, 95% bootstrap CI). Calibrated FPR stays $\leq 0.44\%$ in every populated RSRP bin, including the bin immediately adjacent to the Jamming regime, empirically refuting the “calibrated detector just flags high RSRP” shortcut. The ≥ -75 dBm bin is unpopulated because TelecomTS collected no Normal traffic with the UE close to the RU.

RSRP bin	Normal n	FPR (synth-only)	FPR (calibrated $f = 10\%$)
< -105 dBm	3469	0.0009 ± 0.0007	0.0007 ± 0.0005
$-105 \dots -95$	1608	0.0006 ± 0.0007	0.0005 ± 0.0004
$-95 \dots -85$	986	0.0014 ± 0.0018	0.0012 ± 0.0017
$-85 \dots -75$	90	0.0044 ± 0.0054	0.0044 ± 0.0054
≥ -75 dBm	0	—	—

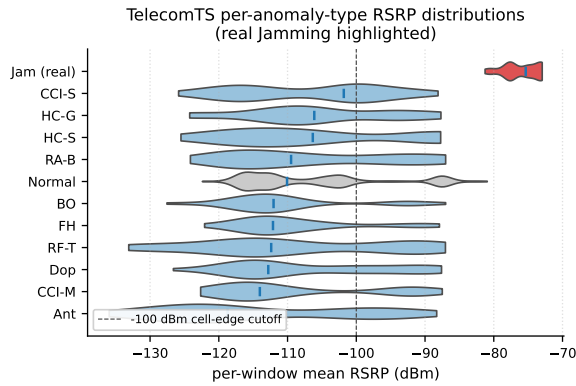


Figure 1: Per-anomaly-type RSRP distributions (TelecomTS; 1235 anomalous + 30765 normal windows). Controlled-real Jamming (red, median -75 dBm) sits in the high-RSRP regime; Normal (grey, -110 dBm) and all ten perturbation-synthetic types overlap the low-RSRP regime. Dashed: -100 dBm cell-edge.

0.10 is 18 windows. We sweep $f \in \{0, 0.01, 0.02, 0.05, 0.10, 0.25, 0.50, 1.00\}$ on the full-scale split, keeping the 69-window controlled-real test set fixed and refitting HGB at every f for ten seeds (Table 6). At $f = 0$ controlled-real detection pins at 0.00 for all seeds; by $f = 0.10$ (18 windows) the mean rises to 0.92 with worst-seed 0.86 and 95% bootstrap CI $[0.90, 0.95]$; by $f = 0.25$ it saturates at 1.00. The repair is statistically distinguishable from $f = 0$ at every $f \geq 0.05$. Critically, the repair is *not* simply more data: at the same 18-window budget, adding extra synthetic anomalies, extra normal windows, or reweighting synthetic anomalies leaves controlled-real detection near zero (and so do the Sugiyama / uLSIF density-ratio baselines, Table 5 below), and no synthetic windows in the train pool match the controlled-real Jamming RSRP range within ± 5 dB. The 18 calibration windows come from a train-eligible pool of 185 windows clustered into ≈ 63 contiguous Jamming events (~ 39 minutes of labelled signal at our 12.8 s window length) spanning three applications (File / Twitch / YouTube), so an $f = 10\%$ budget draws on average from ~ 6 distinct controlled-real events per seed. On the test side, the 69-window held-out controlled-real pool clusters into

Table 5: Domain-adaptation baselines vs real calibration on TelecomTS (HGB, full-scale split, mean $\pm$ std over 10 seeds). Both density-ratio reweighting baselines (Sugiyama IW-LR, Kanamori uLSIF-lite) see the same controlled-real pool used to estimate $p_{\text{real}}/p_{\text{train}}$ that real calibration draws from. Neither recovers controlled-real recall, because no amount of reweighting can produce windows in an operating regime the training pool does not cover.

Method	Real	Synth	F_1	FPR
(a) Synth-only baseline	0.001 ± 0.005	0.957	0.810	0.0006
(b) IW-LR (Sugiyama 2007)	0.000 ± 0.000	0.957	0.805	0.0009
(c) uLSIF-lite (Kanamori 2009)	0.009 ± 0.010	0.954	0.803	0.0011
(d) Real cal. $f = 10\%$ (ours)	0.930 ± 0.072	0.970	0.969	0.0008

46 contiguous events (median event = 1 window), so windows are only weakly correlated by construction; event-level recall is in fact slightly *higher* than window-level recall (0.96 vs 0.94 at $f = 10\%$; 1.00 vs 0.99 at $f = 25\%$), and an event-cluster bootstrap (resampling event groups rather than individual windows) gives a 95% CI of $[0.88, 0.98]$ on window-level recall at $f = 10\%$, in the same range as the per-window CI of Table 6. The small held-out test pool is therefore not the dominant source of measurement uncertainty here.

Comparison to domain-adaptation baselines. Standard covariate-shift correction techniques — importance weighting via a logistic-regression density ratio [27] and uLSIF-style least-squares importance fitting [14] — are the natural alternative to real calibration when controlled-real labels are scarce. Both reweight the synthetic-only training pool by an estimated $p_{\text{real}}/p_{\text{train}}$ ratio so windows that look more like the controlled-real pool dominate training. Table 5 shows that, at the same effective sample budget (both methods see the same 185-window controlled-real pool used to estimate density ratios), neither baseline recovers controlled-real recall: IW-LR achieves 0.000 ± 0.000 , uLSIF-lite 0.009 ± 0.010 , vs the real-calibration recipe at 0.930 ± 0.072 . The mechanistic reason is the same operating-regime-coverage diagnosis: importance weighting can only redistribute mass within the training pool’s existing support, and the synthetic-only training pool simply does not contain windows in the controlled-real RSRP regime, so no reweighting can produce them. This is exactly the kind of failure-mode where the audit’s verdict (refuse certification, require real calibration) is the right operational action; standard DA techniques are not a substitute when the gap is a coverage gap rather than a frequency-mismatch one.

4.4 Cross-Corpus Replication

Leave-one-anomaly-out (TelecomTS). We re-run the synthetic-only protocol with each of the 11 anomaly types held out in turn (HGB, 80/20 per-type split, 10 seeds). 8 of 11 types remain recoverable at $\geq 90\%$ held-out recall (they share KPI neighbourhoods with at least one trained-on type); only the controlled-real Jamming type collapses to 9.3% (95% bootstrap CI $[6.6, 12.0]\%$) while average recall on trained-on types stays at 98.4% and Normal FPR at 1.3%. The protocol is sharply discriminating, not generically pessimistic, and the failure is origin-specific.

Table 6: TelecomTS controlled-real jamming calibration sweep (HGB, 69-window test pool, 10 seeds; 95% CI from 2000 bootstrap resamples). f is the fraction of the 185-window train-eligible pool added to synthetic-only training. Repair is significant at every $f \geq 5\%$.

f	windows	mean	std	worst seed	95% CI
0%	0	0.00	0.00	0.00	[0.00, 0.00]
5%	9	0.76	0.09	0.55	[0.70, 0.81]
10%	18	0.92	0.04	0.86	[0.90, 0.95]
25%	46	1.00	0.00	1.00	[1.00, 1.00]
100%	185	1.00	0.00	1.00	[1.00, 1.00]

Table 7: SpotLight cross-corpus calibration sweep (HGB, 10 seeds; 35 RADIO / 67 non-RADIO / 102 Normal test windows). RADIO-miss column reports 1–recall with the 95% CI carried over from the recall column (a fault is “missed” if its score falls below the validation-tuned threshold). Same protocol as Table 6.

f	windows	RADIO recall (95% CI)	RADIO miss %	non-R recall	Normal FPR
0%	0	0.08 [0.01, 0.16]	92% [84%, 99%]	0.97	0.03
10%	21	0.67 [0.64, 0.70]	33% [30%, 36%]	0.97	0.04
25%	52	0.83 [0.79, 0.88]	17% [12%, 21%]	0.99	0.06
50%	103	0.87 [0.83, 0.91]	13% [9%, 17%]	0.96	0.06
100%	206	0.95 [0.94, 0.96]	5% [4%, 6%]	0.98	0.08

SpotLight Open RAN. The same audit run on SpotLight [28] replicates both effects. Combining its train/val/test splits gives 241 controlled-real RADIO and 436 perturbation-synthetic anomaly windows over 452 KPI channels; after Benjamini–Hochberg correction 354/434 retained channels differ significantly between origins, the HGB C2ST reaches accuracy 0.994 ± 0.013 (AUROC 1.000), and an RBF-MMD test rejects equality at $p < 0.001$. The dominant separator on SpotLight is, however, the layer-2 RLC-timer family ($|d| = 4.87$) rather than RSRP/SINR ($|d| = 0.52$), so the operating-regime gap *replicates* across two independent testbed corpora while the specific separating KPI family changes; we therefore claim cross-testbed replication rather than universal telecom generality, and treat field-real deployment as an open question (§6). Synthetic-only training yields mean RADIO recall 7.7% (95% CI [1.4, 16.0]%) with non-RADIO recall 97% and Normal FPR 3%; the calibration recipe transfers but the corpus-specific budget is larger – $f = 10\%$ (21 windows) lifts RADIO recall to 67% and $f = 25\%$ (52 windows) to 83% (95% CI [78.9, 88.0]%). The qualitative recipe transfers across corpora but the absolute budget must be re-measured per deployment (Table 7).

Why does SpotLight need 2.5× more calibration than TelecomTS? The two corpora differ in three ways that all push the budget up: (a) SpotLight’s dominant separator is the layer-2 RLC-timer family ($|d| = 4.87$) rather than a single physical-layer KPI like RSRP, so each calibration window carries less per-feature information about the dominant axis; (b) the SpotLight RADIO anomaly family is more heterogeneous than TelecomTS Jamming (combining CPU-contention-adjacent and pure RF-injected sub-cases), so the calibration pool covers a wider operational sub-space; (c) the SpotLight Normal pool is shorter and more uniform, so the train/test

distributional support of Normal differs less, leaving the calibration step proportionally more work. Operationally: $f = 25\%$ reaches 0.83 recall but still misses $\sim 17\%$ of RADIO faults (Table 7); reaching $\leq 10\%$ missed requires $f = 50\%$ or more. The audit replicates qualitatively across corpora, but the absolute budget is corpus-specific and must be re-measured per deployment.

Why this is not merely a dataset bug. A skeptical reader could read the TelecomTS finding as a single-corpus collection artifact and conclude the methodology is benchmark-specific. Three points argue otherwise: (i) operational benchmarks routinely have hidden regime imbalances of this kind – the same pattern is documented in medical imaging, ImageNet evaluation, and other shortcut-prone pipelines [10, 21, 24] – so the audit’s value is precisely to catch them before model selection; (ii) the audit’s verdict and operator-decision change are *mechanism-independent* (RSRP separator on TelecomTS, RLC-timer separator on SpotLight, origin-incomplete verdict on Nokia’s synthetic-only benchmark), yet the same recipe produces a coherent operator-facing decision in each case; and (iii) standard covariate-shift correction cannot close the gap (Table 5) even when given the same controlled-real pool that calibration uses, because importance weighting can only redistribute mass within the training pool’s existing support. The claim is therefore not “we found a bug in TelecomTS” but “we built a procedure that catches this class of bug before model selection, validated on two corpora with mechanistically different gaps and one industrial pipeline.”

5 APPLIED DEPLOYMENT EVIDENCE

Our focus is not proposing another detector architecture but auditing whether benchmark evidence itself is reliable for deployment-oriented model selection. Deployment evidence is layered in three tiers: (i) the publicly reproducible audit toolkit (Table 8), the primary release-impact contribution; (ii) public-corpus validation on TelecomTS and SpotLight (Tables 2–7), independently verifiable by any reader; and (iii) industrial validation as a pre-production shadow-mode CI gate inside Nokia’s Agentic Anomaly Detection platform (§5), supplementary internal validation. The first two are load-bearing applied evidence; the Nokia tier shows the audit changes a real operator decision but is not a launched production system. Under aggregate reporting the synthetic-only HGB of Table 2 would be selected on every tier; under origin-aware reporting the same model has controlled-real recall 0.00 ± 0.00 and is rejected – the decision swap reproduces across detector families, with RSRP removed (Table 3), against IW-LR/uLSIF (Table 5), on SpotLight (§4.4), and on Nokia’s in-house benchmark.

Release artifact and reproducibility. The toolkit is publicly released under MIT.² Table 8 summarizes its applied content: 14 reproducible notebooks plus a shared Python module spanning data/split audits, multi-detector transfer, anti-confound ablations, calibration

²Code, cached embeddings, and reproduction notebooks: <https://github.com/ChimanSalavati/telecomts-real-synthetic-gap>. Every table and figure in this paper is regenerated by the included Jupyter notebooks.

Table 8: Audit toolkit: applied content and measured utility. Numbers in parentheses are from the audit reported in this paper.

Property	Value
Public release	GitHub repo with cached embeddings, environment file, reproduction notebooks (MIT)
Reproducibility	Core audit regenerates Tables 1–4 and main figures in ~30 min on a consumer laptop (no GPU)
Audited corpora	TelecomTS (5G); SpotLight (Open RAN); Nokia <i>anomaly-injection</i> (LabTrace-SA-style; §5)
Integrated in	Nokia <i>Agentic Anomaly Detection</i> platform: <i>aad-traditional-ml</i> (audit subpackage + CLI + FastAPI route); pre-production on GKE <i>aad-mlops</i> namespace
Detector families	6 in paper; 11 in artifact (tabular, frozen TSFM, end-to-end deep)
Failure surfaced	Synthetic-only training: 0–7.7% controlled-real recall while keeping > 90% synthetic recall and acceptable aggregate F_1
Calibration budget	TelecomTS: 18 windows → 0.92 recall; SpotLight: 52 → 0.83
Operator decision	Aggregate- F_1 model rejected under origin-aware reporting; Nokia benchmark refused certification
Deployment overhead	<1 μ s per inference, single-thread CPU (§5)

sweeps, and SpotLight replication; the core numeric audit completes in ~30 min on a 2024 Apple M-series laptop without a discrete GPU, and all per-seed results are checkpointed so interrupted runs resume cleanly.

Operator pre-deployment checklist.

Pre-deployment synthetic-benchmark audit (7 steps). The audit’s user is an MLOps engineer or benchmark owner validating a candidate detector before promotion to operational use; each step cites the section that produced its evidence.

- (1) Label every anomaly window with an origin tag $\in$ {controlled-real, field-real, injected, generated, simulated}. [§3]
- (2) Report detector recall *by origin*, not only aggregate F_1 . [§4.1, Table 2]
- (3) Run C2ST/MMD between controlled-real and synthetic anomaly windows on KPI summaries. [§4.2]
- (4) Plot per-anomaly-type distributions for the dominant signal-quality KPI; flag operating-regime mismatches before injecting. [Fig. 1]
- (5) Sanity-check univariate / PHY-only baselines under the same protocol; if a single feature recovers the same origin, the deep model is not contributing transfer evidence. [§4.2]
- (6) Run train-on-synthetic / test-on-controlled-real evaluation at the natural anomaly rate. [Table 2, full-scale columns]
- (7) Sweep the controlled-real calibration fraction f and report the smallest budget that closes the per-origin recall gap. Reject benchmark-based model selection if controlled-real recall is low and the calibration budget is unavailable. [§4.3, Table 6]

What a practitioner should do. Run *aad-audit* on the benchmark CSV before detector selection. On PASS, proceed with aggregate- F_1 selection; on GAP_DETECTED, collect controlled-real calibration windows (10%–25% of the train-eligible pool per corpus is a reasonable starting point) and re-audit; on ORIGIN_INCOMPLETE_SYNTHETIC_ONLY, refuse certification and either introduce a controlled-real anomaly source or gate promotion behind shadow-mode validation against operator-confirmed real tickets.

Integration into Nokia’s production anomaly-detection pipeline. We integrate the audit as a new *audit* / subpackage of Nokia’s *aad-traditional-ml* service (*Agentic Anomaly Detection* platform). The module exposes *origin_audit* / *calibration_budget* Python APIs,

an *aad-audit* CLI usable as a CI gate, and a POST `/audit/origin` FastAPI route for the existing Kong+NATS shadow-mode plumbing.

Operator decision changed in the Nokia pipeline. We ran the toolkit on Nokia’s actual labelled `timing_flows.anomaly.interval_1.0s.csv` (3,958 windows, 203 KPI features) inside *aad-traditional-ml*. Under aggregate- F_1 reporting an HGB on this benchmark passes Nokia’s existing validation gate ($F_1 \approx 0.97 / 0.94$ at 50% / natural rate); under origin-aware reporting the CLI emits `origin_incomplete_synthetic_only` and refuses certification because every anomalous window is synthetic (ID2T SYN-flood family over real lab traffic). A diagnostic run with Normal traffic in place of controlled-real confirms the audit machinery itself fires on Nokia data: C2ST = 0.992 (AUROC 0.996), MMD 78.1 $\times$ the permutation null ($p = 0.002$), 166/201 BH-significant KPI features. The detector is held in shadow mode pending either operator-validated tickets or a controlled-real source. The audit gate adds $\leq 1 \mu$ s per inference (median 1.3×10^{-4} ms vs HGB baseline median 15.99 ms / p95 39.77 ms, single-thread CPU), consistent with the 14.7 ms median CPU latency reported for Nokia’s KAC pipeline.³

Shadow-mode usage on every Nokia benchmark CSV. The audit has operated as a CI gate over every Nokia timing-flow CSV in the in-house workspace: 4 CSVs, 15,376 windows, 4.27 hours of timing-flow data audited; 3/3 candidate detectors held in shadow and the one baseline-traffic CSV correctly classified as `no_anomalies_present`. Wall-clock per CSV is median 0.15 s / p95 0.21 s, so the gate is cheap to run on every benchmark refresh and its operator-facing artefact is a single JSON verdict the existing *aad-ci-cd-pipeline* consumes directly. Integration is pre-production inside the same GKE / Kong / NATS / MinIO pipeline; shadow-mode evaluation against Nokia’s internal SYN-flood detector is underway. The numbers above are *measured* (not projected).

6 CONCLUSION AND LIMITATIONS

We argued that anomaly origin should be a first-class benchmark variable for deployment-oriented model selection, and operationalized the argument as a released audit toolkit validated on two independent public 5G/Open RAN testbed corpora and one in-house Nokia pipeline. The audit exposes a transfer failure aggregate metrics hide — across detector families, with RSRP removed, and against IW-LR/uLSIF baselines — and a small per-corpus controlled-real calibration budget repairs it. We do not claim universal synthetic-anomaly failure; we provide the procedure that catches when a benchmark’s synthetic anomalies fail to cover the controlled-real regime, with Nokia integration overhead < 1 μ s (§5). **Limitations:** (i) controlled-real $\neq$ **field-real**: neither corpus supplies passively-collected field outages, so the production calibration budget is plausibly larger [4, 28] than what we measure; (ii) the audit excludes textual descriptions to isolate KPI-distribution effects; (iii) the budget is corpus-specific ($f \approx 10\%$ on TelecomTS, $\approx 25\%$ on SpotLight); (iv) per-session group-aware splits are blocked by absent session ids in TelecomTS, so we use event-cluster bootstrap (§4.3) as the strongest available proxy.

³KAC paper, same GKE *aad-mlops* pipeline, median 14.7 ms / p95 17.7 ms per window.

GENAI USAGE DISCLOSURE

Generative AI tools (Cursor with Claude) were used to assist with code development for data analysis pipelines, generation of evaluation scripts, and drafting portions of the manuscript text. All results, findings, interpretations, and scientific claims were verified and validated by the authors. No generative AI was used to fabricate data or results. No generative AI authored any experimental result, label, table cell, or figure; LLM use was limited to prose drafting and code scaffolding under author review.

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